

Homework (Habanero)

- Q1.** Solve the system: $x^2 + y^2 = 25$, $x^2 - y^2 = 9$
- (A) $x = pm4, y = pm3$
(B) $x = pm3, y = pm4$
(C) $x = pm2, y = pm5$
(D) $x = pm5, y = pm2$
(E) $x = pm1, y = pm6$
- Q2.** A bacteria culture grows according to $P(t) = 2t^2 + 5t + 1$, where t is time in hours. If $P(2) = 25$, what is $P(4)$?
- (A) 41
(B) 49
(C) 57
(D) 65
(E) 73
- Q3.** The quadratic $x^2 + bx + c$ has roots $x = -2$ and $x = 3$. What is the equation of the quadratic?
- (A) $x^2 + x - 6$
(B) $x^2 - x - 6$
(C) $x^2 + 5x + 6$
(D) $x^2 - 5x + 6$
(E) $x^2 + x + 6$
- Q4.** Solve the system: $y = x^2 - 4$, $y = 2x - 3$
- (A) $x = 1, y = -3$
(B) $x = -1, y = -5$
(C) $x = 2, y = 1$
(D) $x = 3, y = 3$
(E) $x = -3, y = -3$
- Q5.** A quadratic model for the spread of a disease is given by $y = at^2 + bt + c$, where y is the number infected and t is time in days. If $y(1) = 10$ and $y(2) = 25$, what is the equation of the quadratic?
- (A) $y = t^2 + 5t + 4$
(B) $y = t^2 - 5t + 4$
(C) $y = t^2 + t - 4$
(D) $y = t^2 - t + 4$
(E) $y = t^2 + 3t - 4$
- Q6.** What is the x -coordinate of the vertex of the parabola $y = x^2 + 4x + 3$?
- (A) -3
(B) -2
(C) -1
(D) 0
(E) 1
- Q7.** The quadratic $ax^2 + bx + c$ has roots $x = -1$ and $x = 4$. If $a = 2$, what is b ?
- (A) -5
(B) -10
(C) 10
(D) 5
(E) 0
- Q8.** A medical researcher is studying the effect of a new drug on blood pressure. The quadratic model for the change in blood pressure is given by $y = 2x^2 - 5x + 1$, where y is the change in blood pressure and x is the dosage. What is the dosage for a change of -1 ?
- (A) 1
(B) 2
(C) 3
(D) 0.5
(E) 2.5

Open Ended Questions (FRQ)

- Q9.** A hospital is planning to build a new wing. The area of the wing is given by the quadratic $A = x^2 + 4x + 4$, where x is the length of the wing. If the area of the wing must be at least 100 square meters, what is the minimum length of the wing?
- Q10.** A pharmacist is mixing a new medication. The concentration of the medication is given by the quadratic $C = 2t^2 - 5t + 1$, where t is time in minutes. If the concentration must be at least 0.5 milligrams per liter, what is the minimum time required to mix the medication?

Chef's Tips

- Ensure students show all work and explain their reasoning
- Emphasize the importance of checking answers
- Encourage students to use graphs to visualize the solutions